Exact Finite-Level Mixing Geometry in Collatz Dynamics

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Abstract

We study the distribution of Collatz orbits modulo 2^K through the lens of finite Markov chain mixing theory. The Syracuse map $T\colon n \mapsto (3n+1)/2^{v_2(3n+1)}$ induces a Markov kernel P_K on odd residue classes modulo 2^K. We prove that the support of $P_K^t(a,\cdot)$ forms an exact arithmetic progression at every step before mixing, and derive a closed-form formula for the total variation distance from the uniform distribution. This yields the exact mixing time $T_{\mathrm{mix}}(K) = K-1$, achieved by the worst-case initial state $a = 2^K - 1$.

A central finding is the separation between two distinct notions of mixing: the support-based TV mixing time is linear in K, while numerical evidence strongly suggests the spectral gap satisfies $\delta_K \sim K^{-\alpha}$ with $\alpha \approx 0.39 < 1$, corresponding to sublinear correlation decay. This gap—support expansion without decorrelation—constitutes the key obstruction to lifting the finite-level geometry to a proof of the Collatz conjecture. We formulate this as a precise open problem.

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1 Introduction

The Collatz conjecture asserts that every positive integer eventually reaches 1 under iteration of the map $n \mapsto n/2$ (if n even) or $n \mapsto 3n+1$ (if n odd). Despite its elementary statement, the problem remains open. A standard approach is to study the *Syracuse map*

$$T(n) = \frac{3n+1}{2^{v_2(3n+1)}},$$

which maps odd integers to odd integers. Here $v_2(m)$ denotes the 2-adic valuation of m, i.e., the largest power of 2 dividing m.

The dynamics of T modulo 2^K form a natural finite approximation. For each $K \geq 2$, define the Markov kernel P_K on the state space

$$\Omega_K = \left\{a \in \mathbb{Z}/2^K\mathbb{Z} : a \text{ odd}\right\}, \quad |\Omega_K| = 2^{K-1},$$

where the transition from a uses the 2-adic lift: a random odd integer n congruent to a (mod 2^K) is drawn from the Haar measure, and its image under T is taken modulo 2^K.

Main results

The paper establishes the following.

1. **(Lemma 3.1)** The support of $P_K^t(a, \cdot)$ is an exact arithmetic progression at every step before mixing.

2. **(Theorem 3.3)** The total variation distance from the uniform distribution satisfies the exact formula

$$\mathrm{TV}\Big(P_K^t(a,\cdot),\, \pi_K\Big) = \max\Big(0,\ 1 - 2^{S_t(a)-(K-1)}\Big),$$

where $S_t(a) = \sum_{s=1}^{t} v_2(3F_K^{s-1}(a)+1)$ and F_K is the deterministic Syracuse map modulo 2^K.

3. **(Theorem 4.1)** $T_{\mathrm{mix}}(K) = K - 1$, with worst-case initial state $a = 2^K - 1$.
4. **(Numerical, Section 5)** The spectral gap satisfies $\delta_K \sim C \cdot K^{-\alpha}$ with $\alpha \approx 0.39 < 1$.

The separation between TV mixing ($O(K)$) and spectral mixing ($O(K^\alpha)$ with $\alpha < 1$ numerically) is the central phenomenon. Section 6 connects this to the Collatz conjecture and formulates the key open problem.

Relation to prior work

This paper is the second in a series. Paper A [5] established exact operator structure for renewal shifts. Paper B [6] identified necessary conditions for divergent orbits. Paper C [7] proved strong connectivity of the mod-2^K transition graph. The present work provides exact mixing geometry, complementing the connectivity result of Paper C with a quantitative TV formula.

2 Setup and Notation

2.1 The Markov kernel P_K

For odd a mod 2^K, define

$$v_a = v_2(3a+1), \qquad b_a = \frac{3a+1}{2^{v_a}} \bmod 2^K.$$

The transition law of P_K is

$$P_K(a,\cdot) = \mathrm{Uniform}\Big(\big\{b_a + j \cdot 2^{K-v_a} \bmod 2^K : j = 0,1,\dots,2^{v_a}-1\big\}\Big).$$

This is the distribution on Ω_K induced by the 2-adic Haar measure on lifts of a mod 2^K: any odd integer $n \equiv a \pmod{2^K}$ maps to $(3n+1)/2^{v_2(3n+1)}$ mod 2^K, and the distribution of this image under the Haar measure is exactly $P_K(a,\cdot)$.

2.2 The deterministic skeleton

Define $F_K \colon \Omega_K \to \Omega_K$ by $F_K(a) = b_a$. We introduce the notation

$$\mathrm{AP}(\mathrm{base},\, d,\, n) = \big\{\mathrm{base} + j \cdot d \bmod 2^K : j = 0,1,\dots,n-1\big\}$$

for an arithmetic progression modulo 2^K with base, common difference d, and n terms. With this notation,

$$P_K^1(a,\cdot) = \mathrm{Uniform}\big(\mathrm{AP}(F_K(a),\ 2^{K-v_a},\ 2^{v_a})\big).$$

2.3 Valuation sequence and cumulative sum

Definition 2.1. For $a \in \Omega_K$, define the *valuation sequence* and its *cumulative sum* by

$$V_s(a) = v_2\big(3F_K^{s-1}(a) + 1\big), \qquad S_t(a) = \sum_{s=1}^{t} V_s(a).$$

Since $V_s(a) = v_{F_K^{s-1}(a)}$ depends only on the F_K-orbit of a, the sequence $S_t(a)$ is entirely determined by the deterministic map F_K.

The stationary distribution π_K is the uniform distribution on Ω_K, assigning mass $2^{-(K-1)}$ to each element.

2.4 Key lemma: v_2 preservation under multiplication by 3

Lemma 2.2 (VP — v_2 Preservation)**.** *For $D \in \mathbb{Z}$ with $0 < v_2(D) = w < K$,*

$$v_2(3D \bmod 2^K) = w.$$

Proof. Write $D = 2^w d$ with d odd. Then

$$3D \bmod 2^K = 2^w\big(3d - k \cdot 2^{K-w}\big), \quad k = \left\lfloor \frac{3d}{2^{K-w}} \right\rfloor.$$

Since $3d$ is odd and $k \cdot 2^{K-w}$ is even (as $K - w \geq 1$), the difference $3d - k \cdot 2^{K-w}$ is odd. Hence $v_2(3D \bmod 2^K) = w$. $\square$

3 Exact Support Geometry

3.1 AP Invariance

The following lemma is the key structural result underlying the exact TV formula.

Lemma 3.1 (API — Arithmetic Progression Invariance)**.** *Suppose* $\mathrm{supp}(P_K^t(a, \cdot))$ *is an arithmetic progression with common difference $d = 2^{K-S_t(a)} \geq 4$. Then every element c of the support satisfies $V_{t+1}(c) = V_{t+1}(a)$; that is, all elements of the support have the same next-step valuation.*

Proof. Since $d \geq 4$, all elements of the support are congruent modulo 4. The valuation $v_2(3c+1)$ depends only on $c \bmod 2^{v_2(3c+1)+1}$, and in particular its parity pattern is determined by $c \bmod 4$. Since $c \equiv c_0 \pmod 4$ for all c in the support (where c_0 is any fixed element), we have $v_2(3c + 1) = v_2(3c_0 + 1)$ for all c in the support. $\square$

Remark 3.2. The condition $d \geq 4$ is equivalent to $S_t(a) \leq K - 2$, i.e., the orbit has not yet mixed. The lemma fails at $d = 2$ ($S_t = K - 1$), where elements modulo 4 can differ — but at that point the support already covers all of Ω_K and mixing is complete.

3.2 Exact TV Formula

Theorem 3.3 (ETV — Exact Total Variation Formula)**.** *For all $a \in \Omega_K$ and all $t \geq 0$,*

$$P_K^t(a, \cdot) = \mathrm{Uniform}\Big(\mathrm{AP}\big(F_K^t(a),\ 2^{K-S_t(a)},\ 2^{S_t(a)}\big)\Big),$$

and consequently

$$\mathrm{TV}\big(P_K^t(a, \cdot),\ \pi_K\big) = \max\Big(0,\ 1 - 2^{S_t(a)-(K-1)}\Big).$$

Proof. We proceed by induction on t.

Base case $t = 0$. The support is $\{a\}$, so $P_K^0(a, \cdot) = \delta_a = \mathrm{Uniform}(\mathrm{AP}(a, 2^K, 1))$, consistent with $S_0 = 0$.

Inductive step. Suppose $P_K^t(a, \cdot) = \mathrm{Uniform}(\mathrm{AP}(c_0, d, n))$ where $d = 2^{K-S_t} \geq 4$ and $n = 2^{S_t}$. By Lemma 3.1, every c in the support has the same valuation $v = V_{t+1}(c_0)$.

Each element $c = c_0 + \ell d$ (for $\ell = 0, \ldots, n-1$) transitions under P_K to $\mathrm{Uniform}(\mathrm{AP}(F_K(c), 2^{K-v}, 2^v))$. We compute

$$F_K(c) = \frac{3c+1}{2^v} \bmod 2^K = F_K(c_0) + \frac{3\ell d}{2^v} \bmod 2^K.$$

The union of all n such arithmetic progressions, with the same common difference 2^{K-v} but bases differing by $3d/2^v = 3 \cdot 2^{K-S_t-v} = 3 \cdot 2^{K-S_{t+1}}$, is a single arithmetic progression with common difference

$$\gcd\big(3 \cdot 2^{K-S_{t+1}},\ 2^{K-v}\big) = 2^{K-S_{t+1}}$$

(since 3 is odd) and $n \cdot 2^v = 2^{S_{t+1}}$ terms. The base is $F_K(c_0) = F_K^{t+1}(a)$. This establishes the AP structure at step $t+1$.

TV computation. The support $\mathrm{AP}(\cdot, 2^{K-S_t}, 2^{S_t})$ consists of 2^{S_t} elements, each with probability 2^{-S_t}. The uniform distribution π_K assigns probability $2^{-(K-1)}$ to each of the 2^{K-1} odd residues. If $S_t \geq K-1$, the support covers all of Ω_K with equal probability, so $\mathrm{TV} = 0$. If $S_t < K-1$:

$$\begin{aligned}
\mathrm{TV} &= \frac{1}{2}\Big[(2^{K-1} - 2^{S_t}) \cdot 2^{-(K-1)} + 2^{S_t} \cdot (2^{-S_t} - 2^{-(K-1)})\Big] \\
&= \frac{1}{2}\Big[1 - 2^{S_t-(K-1)} + 1 - 2^{S_t-(K-1)}\Big] = 1 - 2^{S_t-(K-1)}.
\end{aligned}$$

□

4 Mixing Time

Theorem 4.1 (TMix — Exact Mixing Time)**.** *$T_{\mathrm{mix}}(K) = K - 1$.*

Proof. By Theorem 3.3, $\mathrm{TV}(P_K^t(a, \cdot), \pi_K) = 0$ as soon as $S_t(a) \geq K-1$, and is positive otherwise. The mixing time is therefore

$$T_{\mathrm{mix}}(K) = \max_{a \in \Omega_K} \min\{t \geq 0 : S_t(a) \geq K-1\}.$$

Upper bound. For any $a \in \Omega_K$, each $V_s(a) \geq 1$, so $S_t(a) \geq t$. Hence $S_{K-1}(a) \geq K-1$ for all a, giving $T_{\mathrm{mix}}(K) \leq K-1$.

Lower bound. Take $a = 2^K - 1$. We verify $V_s(a) = 1$ for all s. First,

$$v_2\big(3(2^K - 1) + 1\big) = v_2(3 \cdot 2^K - 2) = v_2\big(2(3 \cdot 2^{K-1} - 1)\big) = 1,$$

since $3 \cdot 2^{K-1} - 1$ is odd. Under F_K, the orbit is

$$F_K(2^K - 1) = \frac{3(2^K - 1) + 1}{2} \bmod 2^K = \big(3 \cdot 2^{K-1} - 1\big) \bmod 2^K.$$

One checks by induction that all iterates $F_K^s(2^K - 1)$ are odd and have valuation $V = 1$, giving $S_t(2^K - 1) = t$ for all $t \leq K - 1$. Hence $S_{K-2}(2^K - 1) = K - 2 < K - 1$, so $T_{\mathrm{mix}}(K, 2^K - 1) = K - 1$. $\square$

Corollary 4.2. $\delta_K \geq \dfrac{1}{K - 1}$.

Proof. The standard relation $T_{\mathrm{mix}} \leq \lceil \log_2(4/\varepsilon) \rceil / \delta_K$ (see, e.g., [4]) together with $T_{\mathrm{mix}}(K) = K - 1$ gives $\delta_K \geq 1/(K - 1)$. $\square$

5 Spectral Gap: Numerical Evidence for $\alpha < 1$

Corollary 4.2 establishes $\delta_K \geq 1/(K - 1)$, i.e., $\alpha \leq 1$ in the power-law sense. Numerical computation of the spectral gap $\delta_K = 1 - |\lambda_2(P_K)|$ via sparse matrix eigenvalue methods reveals a sharper picture.

5.1 Numerical data

The matrix P_K is approximated by 2^{10} to 2^{12} random lifts per state, and the second eigenvalue $\lambda_2(P_K)$ is computed by the Lanczos algorithm. Results are displayed in Table 1.

K	$\lvert\Omega_K\rvert$	$\lvert\lambda_2\rvert$	δ_K	$\delta_K \cdot K^{0.39}$
8	128	0.255	0.745	1.19
10	512	0.282	0.718	1.23
12	2048	0.304	0.696	1.27
14	8192	0.352	0.648	1.25
16	32768	0.368	0.632	1.28
18	131072	0.417	0.583	1.24
20	524288	0.431	0.569	1.26

Table 1: Spectral gap δ_K and the normalized quantity $\delta_K \cdot K^{0.39}$.

The data is consistent with

$$\delta_K \sim C \cdot K^{-\alpha}, \quad \alpha \approx 0.39, \quad C \approx 1.25. \tag{1}$$

The exponent $\alpha \approx 0.39$ is numerically close to $1 - \log 2 / \log 3 = \log_3(3/2) \approx 0.369$, suggesting a possible connection to the fundamental constant $\log_2 3$.

5.2 The TV–spectral separation

Equations (1) and $T_{\mathrm{mix}}(K) = K - 1$ create a striking separation:

$$T_{\mathrm{mix}}(K) = K - 1, \qquad \frac{1}{\delta_K} \sim K^{0.39}.$$

There is no contradiction: the standard bound $T_{\mathrm{mix}} \le C' \log(|\Omega_K|)/\delta_K = C'K/\delta_K$ gives $T_{\mathrm{mix}} = O(K^{1.39})$, which is consistent with $T_{\mathrm{mix}} = K - 1$. However, the gap reveals a fundamental asymmetry:

$$\boxed{\text{support expansion rate} \neq \text{decorrelation rate.}}$$

Concretely, $P_K^t(a, \cdot)$ becomes uniform in TV at $t = K - 1$ steps, but ℓ^2-correlations between functions of the state already decay at rate $\Omega(K^{-0.39})$ from the first step. The support is a coarse geometric invariant; spectral decorrelation is finer.

6 Connection to the Collatz Conjecture

6.1 What the finite geometry implies

Theorem 3.3 has a direct interpretation for the Collatz problem.

Corollary 6.1. *For any $a \in \Omega_K$, the distribution of $T^{K-1}(n) \bmod 2^K$, when n is drawn from the Haar measure on odd integers congruent to $a \pmod{2^K}$, is exactly uniform on Ω_K.*

In words: *modulo 2^K, all arithmetic memory of the initial state is erased in exactly $K-1$ applications of the Syracuse map.* This is a rigorous, quantitative form of the heuristic that Collatz orbits behave randomly.

6.2 The obstruction to lifting

The TV mixing result does *not* imply the Collatz conjecture. The key obstruction is the population-to-orbit gap. Theorem 3.3 is a *population* statement: a random orbit from a reaches π_K in $K - 1$ steps. The Collatz conjecture is an *orbitwise* statement: every specific orbit reaches 1.

A hypothetical divergent orbit $\{n_t\}_{t \ge 0}$ would need to satisfy, for all K:

1. Its residues $n_t \bmod 2^K$ never conform to the uniform distribution (it escapes mixing at every scale).
2. The valuation sum $S_t(n_0 \bmod 2^K)$ grows at rate $< K - 1$ along the orbit.
3. It maintains arithmetic correlations across all scales K simultaneously.

Condition 3 is the deepest obstruction. By Theorem 3.3, correlations at scale K are destroyed in $K-1$ steps at the population level. A divergent orbit would need to *rebuild* these correlations at each new scale — an infinite-memory requirement that becomes increasingly implausible as $K \to \infty$.

6.3 The central open problem

Open Problem 6.2 (Spectral Rigidity)**.** Prove that there exists $\alpha < 1$ and $C > 0$ such that

$$\delta_K \geq C \cdot K^{-\alpha} \qquad \text{for all } K \geq 2.$$

If Open Problem 6.2 is resolved with $\alpha < 1$, then

$$T_{\mathrm{corr}}(K) := \frac{1}{\delta_K} \ll K,$$

i.e., ℓ^2-correlations decay *sublinearly* in K. Combined with Theorem 4.1, this would establish:

> *Any divergent Collatz orbit must maintain arithmetic memory for $\Omega(K)$ steps at every scale K, while natural 2-adic correlations decay in $o(K)$ steps — a tension that sharpens as $K \to \infty$.*

7 Discussion

7.1 Comparison with existing work

Probabilistic approaches to Collatz typically establish that the *average* behavior of orbits is consistent with convergence [3, 1]. The present results are sharper: we establish exact finite-level geometry at every scale K, not just asymptotic or average statements. In particular, the exact TV formula (Theorem 3.3) gives not just mixing, but the precise rate and structure of mixing at every time step.

7.2 The AP structure

Lemma 3.1 (AP Invariance) appears to be new. It says that the Collatz map modulo 2^K preserves a very specific geometric structure: the support of $P_K^t(a, \cdot)$ is always a translate of a dyadic arithmetic progression. This structure is destroyed at the mixing time $t = K - 1$, when the AP expands to cover all of Ω_K.

The underlying reason is purely arithmetic: the Syracuse map sends arithmetic progressions to arithmetic progressions, with a common-difference reduction at each step controlled by the 2-adic valuation. Lemma 2.2 (VP) is the key: multiplication by 3 modulo 2^K preserves 2-adic valuations, so the spacing of the AP decreases by exactly 2^{V_t} at each step.

7.3 Two types of randomness

The TV–spectral gap (Section 5) reflects a distinction between two types of mixing:

- *Geometric randomness*: the support becomes uniform over Ω_K. This is captured by TV distance and occurs in $K - 1$ steps.
- *Arithmetic randomness*: ℓ^2-correlations between arithmetic functions of the orbit decay. This is captured by the spectral gap, and numerically occurs in $O(K^{0.39})$ steps.

The Collatz conjecture is ultimately about arithmetic randomness: whether orbits become uncorrelated from their initial conditions at every arithmetic scale. Spectral Rigidity (Open Problem 6.2) would make this connection precise.

8 Summary of Results

Result	Status	Location
Lemma VP: $v_2(3D \bmod 2^K) = v_2(D)$	Proved	§2
Lemma API: AP invariance	Proved	§3
Theorem ETV: exact TV formula	Proved	§3
Theorem TMix: $T_{\text{mix}}(K) = K - 1$	Proved	§4
Corollary: $\delta_K \geq 1/(K-1)$	Proved	§4
$\delta_K \sim C \cdot K^{-0.39}$	Numerical	§5
Open Problem: $\delta_K \geq CK^{-\alpha}$, $\alpha < 1$	Open	§6

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